

# ***SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE***

**SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical**

## **PRACTICAL - 2**

**AIM :-** To implement Iterative Deepening Depth-First Search (IDDFS) to find a route from Home (Kush Sapphire) to BKC Complex using the available road network, compare the three available routes based on distance and travel time, and determine the route selected by IDDFS.

### **Code:-**

```
#T101 RAJDEEP M PARAB
# AIM: Implement Iterative Deepening Depth-First Search (IDDFS)
# Problem: Find a route from Home to BKC Complex.
dict_hn = {
    'Home': 0,
    'Mhatarpada Rd': 1,
    'Amit Naik Marg': 2,
    'Ceasar Cross Rd': 3,
    'Jai Bhavani Mata Rd': 4,
    'Ceasar Rd': 5,
    'Swami Vivekanand Rd': 6,
    'BKC': 0
}

# ROAD NETWORK
dict_gn = {

    # Common starting portion
    'Home': {
        'Mhatarpada Rd': 1
    },
    'Mhatarpada Rd': {
        'Amit Naik Marg': 1
    },
    'Amit Naik Marg': {
        'Ceasar Cross Rd': 1
    },
    'Ceasar Cross Rd': {
        'Jai Bhavani Mata Rd': 1
    },
    'Jai Bhavani Mata Rd': {
        'Ceasar Rd': 1
    },
    'Ceasar Rd': {
        'Route 1 - Swami Vivekanand Rd': 1,
        'Route 2 - NH 48': 1,
        'Route 3 - Swami Vivekanand Rd': 1
    },

    # ROUTE 1
    # Distance = 11.7 km
```

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```
'Route 1 - Swami Vivekanand Rd': {  
  'Route 1 - Service Rd': 1  
},  
'Route 1 - Service Rd': {  
  'Route 1 - NH 48': 1  
},  
'Route 1 - NH 48': {  
  'Route 1 - GK Gokhale Bridge': 1  
},  
  
'Route 1 - GK Gokhale Bridge': {  
  'Route 1 - Professor NS Phadke Rd': 1  
},  
'Route 1 - Professor NS Phadke Rd': {  
  'Route 1 - Bharat Nagar Rd': 1  
},  
'Route 1 - Bharat Nagar Rd': {  
  'Route 1 - BKC Rd': 1  
},  
'Route 1 - BKC Rd': {  
  'BKC': 1  
},  
  
# ROUTE 2  
# Distance = 12.2 km  
'Route 2 - NH 48': {  
  'Route 2 - Andheri Subway Rd / Village Rd': 1  
},  
'Route 2 - Andheri Subway Rd / Village Rd': {  
  'Route 2 - Nicolas Wadi Rd / Old Nagardas Rd': 1  
},  
'Route 2 - Nicolas Wadi Rd / Old Nagardas Rd': {  
  'Route 2 - Jijamata Rd / Old Nagardas Rd': 1  
},  
'Route 2 - Jijamata Rd / Old Nagardas Rd': {  
  'Route 2 - Service Rd': 1  
},  
'Route 2 - Service Rd': {  
  'Route 2 - NH 48 Flyover': 1  
},  
'Route 2 - NH 48 Flyover': {  
  'Route 2 - Bharat Nagar Rd': 1  
},  
'Route 2 - Bharat Nagar Rd': {  
  'Route 2 - BKC Rd': 1  
},  
'Route 2 - BKC Rd': {  
  'BKC': 1  
},  
  
# ROUTE 3
```

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```
# Distance = 11.8 km
'Route 3 - Swami Vivekanand Rd': {
  'Route 3 - NH 48': 1
},
'Route 3 - NH 48': {
  'Route 3 - GK Gokhale Bridge': 1
},
'Route 3 - GK Gokhale Bridge': {
  'Route 3 - Professor NS Phadke Rd': 1
},
'Route 3 - Professor NS Phadke Rd': {
  'Route 3 - Jiva Mahale Marg': 1
},
'Route 3 - Jiva Mahale Marg': {
  'Route 3 - Sahar Rd': 1
},
'Route 3 - Sahar Rd': {
  'Route 3 - Service Rd': 1
},
'Route 3 - Service Rd': {
  'Route 3 - NH 48 Flyover': 1
},
'Route 3 - NH 48 Flyover': {
  'Route 3 - Bharat Nagar Rd': 1
},
'Route 3 - Bharat Nagar Rd': {
  'Route 3 - BKC Rd': 1
},
'Route 3 - BKC Rd': {
  'BKC': 1
}
}
```

# ROUTE INFORMATION FROM THE PROVIDED MAPS

```
route_distances = {
  'Route 1': 11.7,
  'Route 2': 12.2,
  'Route 3': 11.8
}
route_times = {
  'Route 1': 29,
  'Route 2': 28,
  'Route 3': 29
}
```

# Complete routes for displaying and comparing

```
routes = {
  'Route 1': [
    'Home',
    'Mhatarpada Rd',
    'Amit Naik Marg',
```

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```
'Ceasar Cross Rd',
'Jai Bhavani Mata Rd',
'Ceasar Rd',
'Swami Vivekanand Rd',
'Service Rd',
'NH 48',
'GK Gokhale Bridge',
'Professor NS Phadke Rd',
'Bharat Nagar Rd',
'BKC Rd',
'BKC'
],
'Route 2': [
'Home',
'Mhatarpada Rd',
'Amit Naik Marg',
'Ceasar Cross Rd',
'Jai Bhavani Mata Rd',
'Ceasar Rd',
'NH 48',
'Andheri Subway Rd / Village Rd',
'Nicolas Wadi Rd / Old Nagardas Rd',
'Jijamata Rd / Old Nagardas Rd',
'Service Rd',
'NH 48 Flyover',
'Bharat Nagar Rd',
'BKC Rd',
'BKC'
],
'Route 3': [
'Home',
'Mhatarpada Rd',
'Amit Naik Marg',
'Ceasar Cross Rd',
'Jai Bhavani Mata Rd',
'Ceasar Rd',
'Swami Vivekanand Rd',
'NH 48',
'GK Gokhale Bridge',
'Professor NS Phadke Rd',
'Jiva Mahale Marg',
'Sahar Rd',
'Service Rd',
'NH 48 Flyover',
'Bharat Nagar Rd',
'BKC Rd',
'BKC'
]
}
start = 'Home'
goal = 'BKC'
```

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```
result = "
```

```
# DEPTH LIMITED SEARCH
```

```
def DLS(city, visitedstack, startlimit, endlimit):
```

```
    global result
```

```
    found = 0
```

```
    result = result + city + ' '
```

```
    visitedstack.append(city)
```

```
    # Goal test
```

```
    if city == goal:
```

```
        return 1
```

```
    # Depth limit reached
```

```
    if startlimit == endlimit:
```

```
        return 0
```

```
    # Explore neighbouring cities/locations
```

```
    for eachcity in dict_gn[city].keys():
```

```
        if eachcity not in visitedstack:
```

```
            found = DLS(
```

```
                eachcity,
```

```
                visitedstack,
```

```
                startlimit + 1,
```

```
                endlimit
```

```
            )
```

```
            if found:
```

```
                return found
```

```
    return 0
```

```
# ITERATIVE DEEPENING DEPTH FIRST SEARCH
```

```
def IDDFS(city, visitedstack, endlimit):
```

```
    global result
```

```
    for i in range(0, endlimit + 1):
```

```
        print("\nSearching at Limit:", i)
```

```
        found = DLS(
```

```
            city,
```

```
            visitedstack,
```

```
            0,
```

```
            i
```

```
        )
```

```
        if found:
```

```
            print("Found")
```

```
            return result
```

```
        else:
```

```
            print("Not Found!")
```

```
            print(result)
```

```
            print("-----")
```

```
            result = "
```

```
            visitedstack.clear()
```

```
    return result
```

```
# DISPLAY ALL THREE ROUTES
```

```
def display_routes():
```

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```
print("AVAILABLE ROUTES FROM HOME TO BKC COMPLEX")
for route_name, route in routes.items():
    print("\n", route_name)
    print("-" * 45)
    print("-> ".join(route))
    print(
        "Distance:",
        route_distances[route_name],
        "km"
    )
    print(
        "Time:",
        route_times[route_name],
        "minutes"
    )
```

```
# COMPARE ROUTES
def compare_routes():
    print("ROUTE COMPARISON")
    for route_name in route_distances:
        print(
            route_name,
            ".",
            route_distances[route_name],
            "km,",
            route_times[route_name],
            "minutes"
        )
    shortest_route = min(
        route_distances,
        key=route_distances.get
    )
    fastest_route = min(
        route_times,
        key=route_times.get
    )
    print("\nShortest Route:")
    print(
        shortest_route,
        "-",
        route_distances[shortest_route],
        "km"
    )
    print("\nFastest Route:")
    print(
        fastest_route,
        "-",
        route_times[fastest_route],
        "minutes"
    )
```

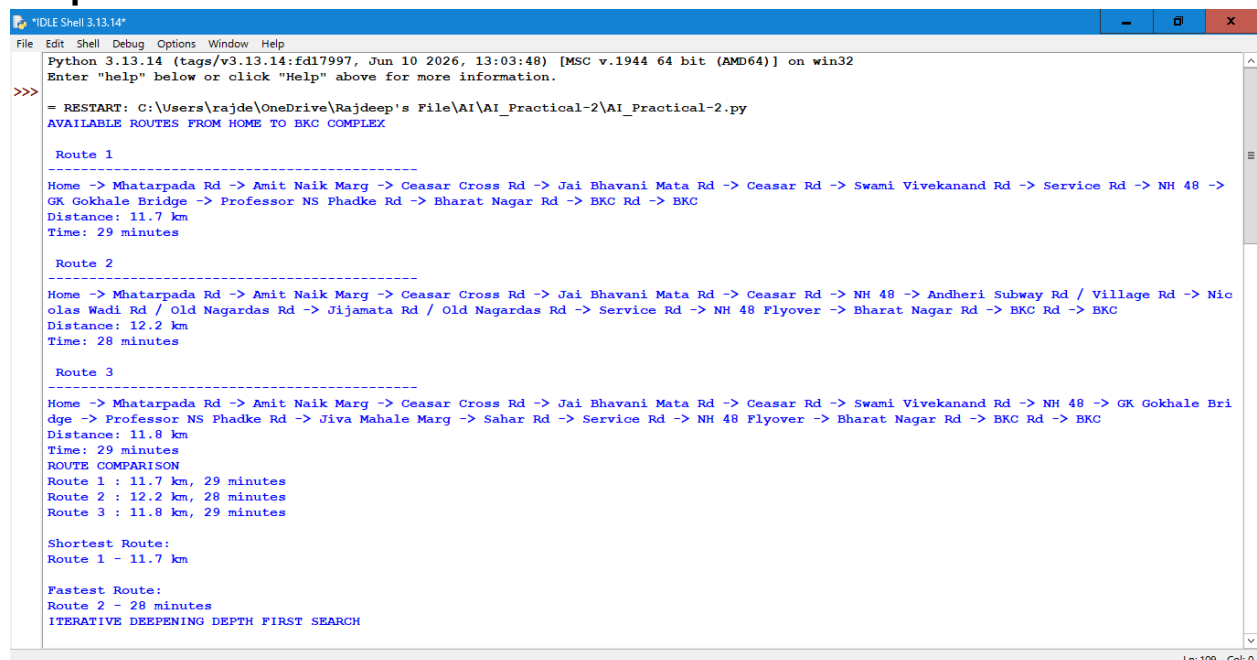
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# MAIN FUNCTION

```
def main():
    global result
    # Display the three routes first
    display_routes()
    # Compare distances and time
    compare_routes()
    # Apply IDDFS
    print("ITERATIVE DEEPENING DEPTH FIRST SEARCH")
    visitedstack = []
    result = ""
    IDDFS(
        start,
        visitedstack,
        20
    )
    print("\nIDDFS Traversal from", start, "to", goal, "is:")
    print(result)
    print("\nFinal IDDFS Route:")
    print(
        result.replace(
            "Route 1 - ", ""
        ).replace(
            "Route 2 - ", ""
        ).replace(
            "Route 3 - ", ""
        )
    )
    main()
```

**Output:-**



```
"IDLE Shell 3.13.14"
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\rajde\OneDrive\Rajdeep's File\AI\AI_Practical-2\AI_Practical-2.py
AVAILABLE ROUTES FROM HOME TO BKC COMPLEX

Route 1
-----
Home -> Mhatarpada Rd -> Amit Naik Marg -> Ceasar Cross Rd -> Jai Bhavani Mata Rd -> Ceasar Rd -> Swami Vivekanand Rd -> Service Rd -> NH 48 -> GK Gokhale Bridge -> Professor NS Phadke Rd -> Bharat Nagar Rd -> BKC Rd -> BKC
Distance: 11.7 km
Time: 29 minutes

Route 2
-----
Home -> Mhatarpada Rd -> Amit Naik Marg -> Ceasar Cross Rd -> Jai Bhavani Mata Rd -> Ceasar Rd -> NH 48 -> Andheri Subway Rd / Village Rd -> Nicolas Wadi Rd / Old Nagardas Rd -> Jijamata Rd / Old Nagardas Rd -> Service Rd -> NH 48 Flyover -> Bharat Nagar Rd -> BKC Rd -> BKC
Distance: 12.2 km
Time: 28 minutes

Route 3
-----
Home -> Mhatarpada Rd -> Amit Naik Marg -> Ceasar Cross Rd -> Jai Bhavani Mata Rd -> Ceasar Rd -> Swami Vivekanand Rd -> NH 48 -> GK Gokhale Bridge -> Professor NS Phadke Rd -> Jiva Mahale Marg -> Sahar Rd -> Service Rd -> NH 48 Flyover -> Bharat Nagar Rd -> BKC Rd -> BKC
Distance: 11.8 km
Time: 29 minutes
ROUTE COMPARISON
Route 1 : 11.7 km, 29 minutes
Route 2 : 12.2 km, 28 minutes
Route 3 : 11.8 km, 29 minutes

Shortest Route:
Route 1 - 11.7 km

Fastest Route:
Route 2 - 28 minutes
ITERATIVE DEEPENING DEPTH FIRST SEARCH
```

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```
*IDLE Shell 3.13.14*
File Edit Shell Debug Options Window Help
-----
Searching at Limit: 10
Not Found!
Home Mhatarpada Rd Amit Naik Marg Ceasar Cross Rd Jai Bhavani Mata Rd Ceasar Rd Route 1 - Swami Vivekanand Rd Route 1 - Service Rd Route 1 - NH
48 Route 1 - GK Gokhale Bridge Route 1 - Professor NS Phadke Rd Route 2 - NH 48 Route 2 - Andheri Subway Rd / Village Rd Route 2 - Nicolas Wadi
Rd / Old Nagardas Rd Route 2 - Jijamata Rd / Old Nagardas Rd Route 2 - Service Rd Route 3 - Swami Vivekanand Rd Route 3 - NH 48 Route 3 - GK Gok
hale Bridge Route 3 - Professor NS Phadke Rd Route 3 - Jiva Mahale Marg
-----
Searching at Limit: 11
Not Found!
Home Mhatarpada Rd Amit Naik Marg Ceasar Cross Rd Jai Bhavani Mata Rd Ceasar Rd Route 1 - Swami Vivekanand Rd Route 1 - Service Rd Route 1 - NH
48 Route 1 - GK Gokhale Bridge Route 1 - Professor NS Phadke Rd Route 1 - Bharat Nagar Rd Route 2 - NH 48 Route 2 - Andheri Subway Rd / Village
Rd Route 2 - Nicolas Wadi Rd / Old Nagardas Rd Route 2 - Jijamata Rd / Old Nagardas Rd Route 2 - Service Rd Route 2 - NH 48 Flyover Route 3 - Sw
ami Vivekanand Rd Route 3 - NH 48 Route 3 - GK Gokhale Bridge Route 3 - Professor NS Phadke Rd Route 3 - Jiva Mahale Marg Route 3 - Sahar Rd
-----
Searching at Limit: 12
Not Found!
Home Mhatarpada Rd Amit Naik Marg Ceasar Cross Rd Jai Bhavani Mata Rd Ceasar Rd Route 1 - Swami Vivekanand Rd Route 1 - Service Rd Route 1 - NH
48 Route 1 - GK Gokhale Bridge Route 1 - Professor NS Phadke Rd Route 1 - Bharat Nagar Rd Route 1 - BKC Rd Route 2 - NH 48 Route 2 - Andheri Sub
way Rd / Village Rd Route 2 - Nicolas Wadi Rd / Old Nagardas Rd Route 2 - Jijamata Rd / Old Nagardas Rd Route 2 - Service Rd Route 2 - NH 48 Fly
over Route 2 - Bharat Nagar Rd Route 3 - Swami Vivekanand Rd Route 3 - NH 48 Route 3 - GK Gokhale Bridge Route 3 - Professor NS Phadke Rd Route
3 - Jiva Mahale Marg Route 3 - Sahar Rd Route 3 - Service Rd
-----
Searching at Limit: 13
Found
IDDFS Traversal from Home to BKC is:
Home Mhatarpada Rd Amit Naik Marg Ceasar Cross Rd Jai Bhavani Mata Rd Ceasar Rd Route 1 - Swami Vivekanand Rd Route 1 - Service Rd Route 1 - NH
48 Route 1 - GK Gokhale Bridge Route 1 - Professor NS Phadke Rd Route 1 - Bharat Nagar Rd Route 1 - BKC Rd BKC
Final IDDFS Route:
Home Mhatarpada Rd Amit Naik Marg Ceasar Cross Rd Jai Bhavani Mata Rd Ceasar Rd Swami Vivekanand Rd Service Rd NH 48 GK Gokhale Bridge Professor
NS Phadke Rd Bharat Nagar Rd BKC Rd BKC
>>>
```